ANY FPPF COVERING IS A PH COVERING

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Editorial note. No omission markers in-span. The two same-chapter inputs are not carded (one Omitted+Hint, one 'clear from the definitions') and are declared external.

Allowed standard prerequisites (editorial).

- Refinement of fppf coverings by schemes. lemma-refine-fppf-schemes (same chapter, NOT carded, declared external). Replaces the pulled fppf covering by one with scheme members.
- Scheme-level fppf implies ph. More on Morphisms, lemma fppf-ph (cross-chapter, named). Refines the scheme covering by a standard ph covering.
- A-fortiori chain for the smaller topologies. lemma-zariski-etale-smooth-syntomic-fppf (same chapter, NOT carded, declared external). Gives the syntomic/smooth/etale/Zariski corollary.

Source reference. Original lemma 042T: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 1033–1033. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0DFC: sources/stacks-spaces-topologies/pinned-original-source.tex, source lines 998–998. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DBT: focused-seeds/geometry/S9-topo-zariski-etale-smooth-syntomic-fppf-ph/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Any fppf covering is a ph covering, and a fortiori, any syntomic, smooth, étale or Zariski covering is a ph covering.*

Proof. We will show that an fppf covering is a ph covering, and then the rest follows from Lemma 0DFC. Let $\{X_i \to X\}_{i\in I}$ be an fppf covering of algebraic spaces over a base scheme S. Let U be an affine scheme and let $U \to X$ be a morphism. We can refine the fppf covering $\{X_i \times_U U \to U\}_{i\in I}$ by an fppf covering $\{T_i \to U\}_{i\in I}$ where T_i is a scheme (Lemma 042T). Then we can find a standard ph covering $\{U_j \to U\}_{j=1,\ldots,m}$ refining $\{T_i \to U\}_{i\in I}$ by More on Morphisms, Lemma

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0DBT (and the definition of ph coverings for schemes). Thus $\{X_i \to X\}_{i \in I}$ is a ph covering by definition. $\square$